CS24305 — THEORY (3.0 CR) | FULL COURSE MASTER | BIT MESRA | NEP SCHEME

Data Communication & Computer Networks (CS24305) — Full Course Master Book

Complete End-to-End B.Tech CSE 5th Semester Study Book & PYQ Bank

EXECUTIVE MASTER STUDY GUIDE & PROTOCOL BANK

Data Communication & Computer Networks (CS24305)

Birla Institute of Technology, Mesra | B.Tech CSE 5th Semester (NEP 2024–25 Scheme)

📚 Complete Course Structure & Protocol Matrix

Module	Layer & Focus	Key Formulations, RFCs & Protocols
Module I	Physical Layer & Signals	Nyquist & Shannon Capacity, Transmission Impairments, Line Coding (Manchester, AMI), Multiplexing (FDM, TDM, WDM)
Module II	Data Link Layer & MAC	Framing, CRC-32 Polynomial Derivations, Sliding Window ARQs (Go-Back-N, Selective Repeat), CSMA/CD & CSMA/CA
Module III	Network Layer & Routing	IPv4/IPv6 Header Analysis, CIDR Subnetting Calculations, Dijkstra Shortest Path, Bellman-Ford, OSPF, BGP-4
Module IV	Transport Layer Protocols	TCP 3-Way Handshake, TCP State Machine, AIMD Congestion Control, Slow Start, Fast Retransmit, UDP Sockets
Module V	Application Layer & Security	DNS Hierarchical Resolution, HTTP/1.1 vs HTTP/2/3, TLS 1.3 Handshake, RSA Asymmetric Encryption, AES

🎯 Exam Preparation & High-Yield Strategy

This publication-grade master book consolidates all 5 modules with formal mathematical channel capacity proofs, step-by-step worked subnetting & CRC numericals, packet header diagrams, and model answers to BIT Mesra end-semester examination questions.

Module I: Data Communications & Networking Overview — Topics Covered

1. Five Components of Data Communication
2. Network Topologies (Mesh, Star, Bus, Ring)
3. OSI 7-Layer vs. TCP/IP 4-Layer Architecture
4. Transmission Impairments (Attenuation, Delay, Noise)
5. Nyquist Maximum Data Rate Theorem
6. Shannon Channel Capacity Theorem & SNR (dB)

1. Layered Network Models: OSI vs. TCP/IP

OSI Layer	Protocol Data Unit (PDU)	Key Responsibilities	TCP/IP Equivalent
7. Application	Messages / Data	Network virtual terminal, HTTP, FTP, SMTP, DNS	**Application Layer**
6. Presentation	Formatted Data	Translation, Encryption (SSL/TLS), Compression	
5. Session	Dialog Tokens	Session establishment, Dialog control, Sync checkpoints	
4. Transport	Segments (TCP) / Datagrams (UDP)	Port addressing, Segmentation, Flow/Error/Congestion control	**Transport Layer**
3. Network	Packets	Logical IP addressing, Subnet routing, Path determination	**Internet Layer**
2. Data Link	Frames	Physical MAC addressing, Framing, Error detection (CRC)	**Network Access Layer**
1. Physical	Bits	Bit transmission over copper/fiber/radio, signal levels	

2. Channel Capacity Theorems

2.1 Nyquist Theorem (Noiseless Channel)

$$C = 2B\log_2(M) \text{ bps}$$

where B is channel bandwidth in Hertz and M is the number of discrete signal voltage levels.

2.2 Shannon Theorem (Noisy Channel)

$$C = B\log_2(1 + \text{SNR}) \text{ bps}$$

where $\text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}}$. Given $\text{SNR}_{\text{dB}} = 10\log_{10}(\text{SNR}) \implies \text{SNR} = 10^{(\text{SNR}_{\text{dB}}/10)}$.

🏛 **BIT Mesra Mid-Sem Exam Numerical (8 Marks)**

Problem: A telephone line has a bandwidth of 4 kHz and a signal-to-noise ratio of 30 dB. Calculate the theoretical Shannon maximum channel capacity.

Solution:

$$\text{SNR}_{\text{dB}} = 30 \implies 10\log_{10}(\text{SNR}) = 30 \implies \text{SNR} = 10^3 = 1000$$

$$C = B\log_2(1 + \text{SNR}) = 4000 \times \log_2(1001) \approx 4000 \times 9.967 = \mathbf{39,869} \text{ bps} \approx \mathbf{39.87} \text{ kbps}$$

Module II: Transmission Media & Signal Encoding — Topics Covered

1. Guided Media (Twisted Pair, Coax, Optical Fiber)
2. Digital Signal Encoding: NRZ, Manchester, AMI
3. Analog Modulation: ASK, FSK, PSK, QAM
4. Pulse Code Modulation (PCM) & Quantization

1. Digital Signal Encoding Techniques

Encoding Scheme	Rule / Signal Transition	Key Advantages & Drawbacks
NRZ-L (Level)	Bit 0 = High level, Bit 1 = Low level	Simple; susceptible to DC baseline wander and clock drift on long runs of 0s/1s.
NRZI (Invert)	Bit 1 = Transition at start; Bit 0 = No transition	Solves sync for consecutive 1s, but fails on strings of 0s.
Bipolar-AMI	Bit 0 = Zero voltage; Bit 1 = Alternating +V and -V	Zero DC component; easy error detection; fails sync on consecutive 0s.
Manchester (Ethernet)	Bit 0 = High-to-Low transition; Bit 1 = Low-to-High transition	**Self-clocking** (transition at middle of every bit); requires $2\times$ bandwidth.
Diff. Manchester	Transition in middle; Bit 0 = Transition at start; Bit 1 = No transition at start	Differential noise immunity; self-clocking; standard in Token Ring (IEEE 802.5).

2. Analog Modulation & 16-QAM Constellation

Quadrature Amplitude Modulation (QAM) modulates both amplitude and phase simultaneously. In 16-QAM, each transmitted constellation point represents **4 bits** ($\log_2(16) = 4$), enabling high spectral efficiency.

Module III: Error Control & Data Link Protocols — Topics Covered

1. Cyclic Redundancy Check (CRC-16/32 Modulo-2)
2. Hamming Distance & (7, 4) Error Correction
3. Sliding Window ARQ: Stop-and-Wait, GBN, SR
4. HDLC Protocol & Bit Stuffing Mechanics
5. Frequency and Time Division Multiplexing (FDM/TDM)

1. Cyclic Redundancy Check (CRC)

Given data word $D(x)$ of k bits and generator polynomial $G(x)$ of degree n:

1. Append n zero bits to $D(x)$ to form dividend $D(x) \cdot 2^n$.
2. Perform binary Modulo-2 division (XOR) by generator polynomial $G(x)$.
3. The n-bit remainder $R(x)$ is the Frame Check Sequence (FCS). Transmit $T(x) = D(x) \cdot 2^n \oplus R(x)$.

2. Sliding Window ARQ Protocols Comparison

Protocol	Sender Window (W_s)	Receiver Window (W_r)	Efficiency (η)	Retransmission Behavior
Stop-and-Wait	1	1	$\frac{1}{1+2a}$ where $a = \frac{T_{\text{prop}}}{T_{\text{trans}}}$	Retransmits single frame upon timer expiry.
Go-Back-N (GBN)	$2^k - 1$	1	$\frac{W_s}{1+2a}$	Discards out-of-order frames; retransmits entire window from lost frame.
Selective Repeat	2^{k-1}	2^{k-1}	$\frac{W_s}{1+2a}$	Buffers out-of-order frames; retransmits **only the damaged frame** (NAK).

Module IV: Switching, Cellular Networks & LANs — Topics Covered

1. Circuit Switching vs. Packet Switching
2. Cellular Frequency Reuse ($N = i^2 + ij + j^2$)
3. IEEE 802.3 Ethernet & CSMA/CD Backoff
4. IEEE 802.11 Wi-Fi & CSMA/CA (RTS/CTS)
5. Virtual LANs (VLANs & IEEE 802.1Q)

1. Ethernet CSMA/CD vs. Wi-Fi CSMA/CA

Feature	IEEE 802.3 Ethernet (CSMA/CD)	IEEE 802.11 Wi-Fi (CSMA/CA)
Mechanism	Carrier Sense Multiple Access with **Collision Detection**	Carrier Sense Multiple Access with **Collision Avoidance**
Wireless Challenge	Transceiver can detect collision while sending.	Cannot detect collision while transmitting (RF blindness) $\implies$ Must avoid collisions.
Collision Recovery	Emits 32-bit Jam signal; Binary Exponential Backoff ($[0, 2^k - 1] \times 51.2\mu s$).	Interframe Spaces (DIFS, SIFS); RTS/CTS handshaking resolves hidden terminals.

Module V: Internetworking, TCP/IP & Routing — Topics Covered

1. IPv4 Addressing, Classless CIDR & VLSM Subnetting
2. TCP 3-Way Handshake & Congestion Control
3. Dijkstra Shortest Path (OSPF) & Bellman-Ford (RIP)
4. Application Protocols: DNS, DHCP, HTTP/2, SMTP

1. TCP Congestion Control Mechanics

TCP maintains a congestion window ($cwnd$) and slow-start threshold ($ssthresh$):

- **Slow Start:** $cwnd$ starts at 1 MSS and doubles every RTT ($cwnd = cwnd \times 2$) exponentially until $cwnd \geq ssthresh$.
- **Congestion Avoidance:** $cwnd$ increases linearly by 1 MSS per RTT (Additive Increase).
- **Triple Duplicate ACKs (Fast Retransmit & Recovery):** $ssthresh = cwnd/2, cwnd = ssthresh + 3$ MSS, retransmit lost segment immediately.
- **Timeout:** $ssthresh = cwnd/2, cwnd = 1$ MSS, re-enter Slow Start.

🏛 **BIT Mesra VLSM Subnetting Problem (10 Marks)**

Problem: Given network block `192.168.1.0/24`, design subnets for Dept A (100 hosts), Dept B (50 hosts), and Dept C (25 hosts).

Solution:

- **Dept A (100 hosts):** Needs $2^7 - 2 = 126$ IPs $\implies$ $/25$. Subnet: `192.168.1.0/25` (Range: `.1` to `.126`, Broadcast: `.127`).
- **Dept B (50 hosts):** Needs $2^6 - 2 = 62$ IPs $\implies$ $/26$. Subnet: `192.168.1.128/26` (Range: `.129` to `.190`, Broadcast: `.191`).
- **Dept C (25 hosts):** Needs $2^5 - 2 = 30$ IPs $\implies$ $/27$. Subnet: `192.168.1.192/27` (Range: `.193` to `.222`, Broadcast: `.223`).